Orthemology notation gallery

Status: DRAFT — not peer reviewed. Rendered by the R7B mathematical- typesetting pipeline (Decision 0023). Every normative symbol in `docs/notation-registry.yaml` appears below as real mathematics: GitHub renders the `$...$` inline math and fenced `math` blocks through MathJax, and `scripts/build_pdfs.py` renders `artifacts/notation-gallery.pdf` through the pinned Typst compiler via `scripts/latex_to_typst_math.py`. This is a **typography layer, not a notation redesign**: each row's meaning is the registry role, unchanged (Decision 0005). `scripts/validate_math_source.py` asserts a bijection between non-retired symbols and this gallery and that `scripts/validate_pdf_math.py` finds no missing-glyph (notdef/tofu) in the rendered PDF — the defect reproduced in `docs/project-closure/r7b/R7B-PDF-MATH-BASELINE.md` (the core `mu`-vector and `C`-vector rendering as notdef under the old `#raw` path). Below they compose correctly as $\vec{\mu}$ and $\vec{C}$.

1. Normative symbols rendered as mathematics

Registry key: `A`
Rendered: A
Role: declared analysis (identifier + version)

Registry key: `alpha, beta, ...`
Rendered: α, β
Role: actor indices

Registry key: `T = task(A)`
Rendered: $T = \mathrm{task}(A)$
Role: task component of the analysis

Registry key: `M_univ, O_univ`
Rendered: $\mathcal{M}, \mathcal{O}$
Role: universal occurrence and ortheme universes

Registry key: `M_A, O_A`
Rendered: $\mathcal{M}_A \subseteq \mathcal{M},\ \mathcal{O}_A \subseteq \mathcal{O}$
Role: analysis-active domain and repertoire

Registry key: `Inst_A`
Rendered: $\mathrm{Inst}_A \subseteq \mathcal{M}_A \times \mathcal{O}_A$
Role: analysis-relative instantiation (the primitive)

Registry key: `O*(m; A)`
Rendered: $O^*(m; A)$
Role: true (actual) profile of m under A

Registry key: `Pi_A`
Rendered: Π_A
Role: complete-profile space under A

Registry key: `Pi_A_partial`
Rendered: Π_A^∂
Role: partial-profile space under A

Registry key: `C_hat`
Rendered: $\hat{C}_{A,\alpha,t}(m)$
Role: candidate set of complete profiles

Registry key: `p_hat`
Rendered: $\hat{p}_{A,\alpha,t}(m) \in \Pi_A^\partial$
Role: inferred (partial) profile — belief, never ground truth

Registry key: `Omega`
Rendered: $\Omega_{A,\alpha,t}$
Role: observation map

Registry key: `mu`
Rendered: $\mu = \langle g, S_\mu, \mathrm{select}_\mu, \mathrm{prov}, \mathrm{ver} \rangle$
Role: metaortheme type

Registry key: `pi_mu`
Rendered: π_μ
Role: paired meta-policy

Registry key: `mu_bar`
Rendered: $\bar{\mu}_{e,j}$
Role: the j-th metaorthemma in episode e

Registry key: `MetaInst`
Rendered: $\mathrm{MetaInst}(\bar{\mu}_{e,j}, \mu)$
Role: token typing of a metaorthemma under a metaortheme

Registry key: `Gamma_E`
Rendered: $\Gamma_E = (E, \rightsquigarrow)$
Role: episode DAG

Registry key: `e_Gamma`
Rendered: $e_\Gamma = \mathrm{comp}(\Gamma_E)$
Role: composed boundary-level episode

Registry key: `GoalSchema`
Rendered: $\mathrm{GoalSchema}(\alpha)$
Role: parametric goal schema

Registry key: `G_target`
Rendered: $\mathcal{G}_{\alpha,A_\alpha}$
Role: actor-alpha's grounded target profile set

Registry key: `Rep_A`
Rendered: Rep_A
Role: representation family under A

Registry key: `chi`
Rendered: $\chi \in \mathrm{Rep}_A$
Role: one representation

Registry key: `L_A_star`
Rendered: $L^*_A(\chi)$
Role: best attainable loss/risk under representation chi

Registry key: `select_mu`
Rendered: select_μ
Role: metaortheme state-selecting evidence procedure

Registry key: `estatus_e`
Rendered: $\mathrm{estatus}_e$
Role: per-claim evidence-status map of episode e

Registry key: `eps_A`
Rendered: ϵ_A
Role: accepted tolerance of the analysis

Registry key: `theta_stop`
Rendered: θ_{stop}
Role: ANDON stop/escalation risk threshold

Registry key: `K_A`
Rendered: $\mathcal{K}_A$
Role: cause repertoire

Registry key: `R_A`
Rendered: $\mathcal{R}_A$
Role: route repertoire

Registry key: `W_A`
Rendered: $\mathcal{W}_A$
Role: warrant-state repertoire

Registry key: `Q_e`
Rendered: $\mathcal{Q}_e$
Role: claim ledger of episode e

Registry key: `delta_e`
Rendered: δ_e
Role: residual-disposition map of episode e

Registry key: `ReqPath`
Rendered: $\mathrm{ReqPath}(e)$
Role: governance-derived required pathway verdicts

2. Structural constructs (typesetting showcase)

The pipeline handles the full range of publication constructs the corpus uses. These are the constructs named in the Decision 0023 design spike.

2.1 Episode signature (display)

The formal-core episode signature — the exact site of the reproduced notdef defect — rendered as real mathematics (the vectors $\vec{\mu}$, $\vec{C}$ compose):

$$\begin{multline}e = \langle \mathrm{id};\ m,\kappa,v;\ x,H;\ \alpha,w,A,T,\\ t;\ \vec{\mu},\mathrm{MetaTok},\pi;\ \vec{C},\hat{p};\ r;\\ \mathrm{estatus};\ \mathcal{Q};\ \delta;\\ \mathrm{hand\_in},\mathrm{hand\_out};\ a,\\ \mathrm{Succ}\rangle\end{multline}$$

2.2 Multi-line aligned equations

$$\begin{aligned}\mathrm{TokenAdequate}(\bar{\mu},e) &\Leftrightarrow \mathrm{MetaInst}(\bar{\mu},\mu)\\ &\quad\wedge \mathrm{Compatible}(\bar{\mu},A(e))\\ \mathrm{V3c}(e) &\Leftrightarrow\\ &\quad\forall\bar{\mu}\in\mathrm{MetaTok}(e):\\ &\quad\mathrm{TokenAdequate}(\bar{\mu},e)\end{aligned}$$

2.3 Set comprehension and predicate

Grounded target (a set of **profiles**, $\subseteq \Pi_{A_\alpha}$, not of occurrences): $\mathcal{G}_{\alpha,A_\alpha} = \{\, p \in \Pi_{A_\alpha} \mid \mathrm{GoalSchema}(\alpha)(p)\,\}$, and strict soundness $\mathrm{StrictlySoundReasoning}_\chi(e) := \mathrm{ReasoningPathAdequate}_\chi(e) \wedge \mathrm{TokenTruthLinked}_\chi(e)$.

2.4 Underbrace

$$\underbrace{O^*(m;A)}_{\text{ground truth, never asserted from discourse}} \neq \underbrace{\hat{p}_{A,\alpha,t}(m)}_{\text{inferred belief}}$$

2.5 Labeled arrows (world transition vs learner update)

$m_t \xrightarrow{a_t} m_{t+1}$ (occurrence lineage) is not $\theta_t \xrightarrow{U_A(e_t)} \theta_{t+1}$ (learner update).

2.6 Table containing inline math

Object: observation map
Symbol: $\Omega_{A,\alpha,t}$
Non-claim: not the occurrence

Object: inferred profile
Symbol: $\hat{p}_{A,\alpha,t}(m)$
Non-claim: not $O^*(m;A)$

Object: metaorthemma
Symbol: $\bar{\mu}_{e,j}$
Non-claim: not the metaortheme type μ

All notation meanings are those of `docs/notation-registry.yaml`; nothing here introduces a new symbol or claim.